Modification of Statistical Metric Biases in Large-Region and Long-Time-Series Landsat Dataset Due to Insufficient Observations

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***Abstract*— Landsat time series, as the longest fine resolution dataset, has the most significant limitation of relatively low temporal frequency (16 days). However, the presence of clouds, cloud shadows, snow, and the failure of the sensor further reduces the amount of clear data, resulting in insufficient observations, which can easily lead to biases in statistical metrics (i.e., maximum, mean, and percentiles). In this study, we took advantage of Google Earth Engine (GEE) and proposed a Statistical Time-series biAs Modification Model (STAMM) that can generate real Landsat statistical metrics (i.e., Landsat statistical metrics based on sufficient observations) in large regions. STAMM can also quantitatively evaluate the bias of statistical metrics due to insufficient observations. The results show that the original Landsat NDVI 75th percentiles over both the northeast and northwest of China overestimated about 10%–30% compared with real Landsat percentiles, while in the southwest of China, it underestimated about 10%. This issue has persisted for the past 20 years. As for other percentiles, the amount of bias depends on the data distribution. The bias of the 95th percentile is relatively small when there are more clear observations in the high-value period. However, at this time, the bias of the 50th percentile bias is relatively large. Taking the Sentinel percentiles as references, STAMM effectively improves the accuracy of Landsat percentiles (i.e., the root-mean-square error (RMSE) decreased from 0.065 to 0.045). It is supposed to be able to provide consistent and accurate Landsat percentiles in large region and long time series for better studying the interannual change of Earth's surface.**

***Index Terms*— Google Earth Engine (GEE), Landsat, large region, long time series, percentile bias, statistical metrics.**

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I. Introduction

THE Landsat satellite series, considering its long history and relatively high spatial resolution, has been used to study the state of the Earth's surface and its changes for decades [1], [2], [3], especially after it became publicly available in 2008 [3], [4], [5]. Until now, there are still many studies using the time-series dataset of Landsat for accurate land cover mapping, change detection, phenology, as well as forest and land disturbance [3], [6], [7], [8].

However, the most significant limitation of the Landsat dataset is its relatively low temporal frequency (16-day revisit capability [9]). In addition, the presence of cloud, cloud shadow, and snow as well as the failure of the Enhanced Thematic Mapper Plus (ETM+) Scan Line Corrector (SLC) that occurred in May 2003 further reduced the number of clear Landsat observations [9], [10]. Therefore, it is very difficult to find entire clear Landsat images for a specified period. Meanwhile, the time distribution of clear Landsat observations is different in different regions. When using multiple Landsat satellite data together (e.g., Landsat 4, 5, 7, and 8), affected by orbit differences and tile overlaps (Fig. 1), these differences in the spatial and temporal distribution of clear Landsat observations are greater, which will significantly influence the use of time-series data.

For obtaining complete clear Landsat images, image compositing algorithms become common solutions, for example, fusing Landsat images with other high temporal resolution images [11], [12], [13], [14], [15] or calculating statistical metrics (i.e., extreme values, individual ranks, means, percentiles, and regression slopes) from the Landsat time series [16], [17], [18], [19], [20], [21]. Both algorithms can provide clear images; however, they highly depend on the quantity of clear Landsat images that are available. For example, fusing Landsat data with daily Moderate Resolution Imaging Spectroradiometer (MODIS) data provides a representation of the fusion algorithm [11], [12], [13], [14], [15], while this kind of algorithm depends on a few clear Landsat images to make up observation pairs as their input, which for some places with lots of clouds may take a few years [22]. Besides, it is reported to be less effective for places where land cover changed [23] and is difficult to apply to large-region or even global studies.

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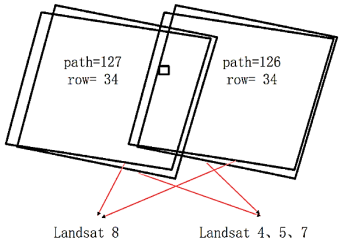


Fig. 1. Tiles overlap of Landsat 4, 5, 7, and 8.

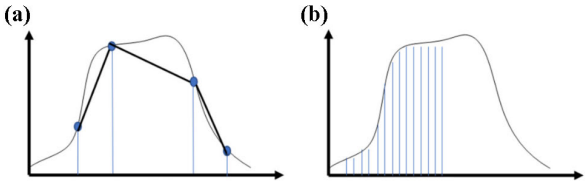


Fig. 2. Impact of Landsat insufficient observations on annual statistical metrics of vegetation. (a) and (b) Two types of insufficient observation distributions that are focused on herein. The gray curve represents the annual change of vegetation index, and the blue vertical lines mark Landsat observations.

The statistical metrics algorithm calculates extreme values, individual ranks, means, percentiles, regression slopes, and other statistical indicators based on the time-series dataset. It is simple but very useful for land cover mapping and change detection methods in large areas [24], [25], [26], [27], [28], [29]. Annual statistical metrics can be used as important landmark features to distinguish different land cover types [8], [25], [30], [31], [32]. In this way, seasonal differences caused by solar angle differences and vegetation phenology changes are minimized, and the data volume is reduced substantially [3]. However, the problem that the time distribution of clear Landsat observations obtained in different regions is different seriously affects the accuracy of statistical metrics. Taking vegetation as an example, the phenology periods reflected by the statistical metrics in different regions and different periods in the same region are not the same. Therefore, due to the insufficient distribution of Landsat's clear observations (Fig. 2), the annual statistical metrics may not fully reflect the vegetation conditions and annual changes, with unknown uncertainties. This uncertainty can lead to errors in land cover mapping, bias in the analysis of vegetation trends [33], [34], and uncertainty in other applications that use the Landsat statistical metrics. At present, there is no quantitative evaluation of the amount of this uncertainty and its impact on subsequent applications.

Among all the statistical metrics, the annual percentile is relatively robust to noise and contains certain time information, which is often used in land cover mapping and vegetation change research [3]. The 50th percentile is close to the mean value, which can reflect the mid-growth condition of the vegetation, while the 95th percentile is close to the maximum value, reflecting the lushest vegetation, and the outliers can also be excluded. However, when there is uncertainty caused by insufficient observations, there will be a significant bias in the statistically obtained percentile [33]. For example, due to the influence of the rainy season, there are few clear observations during the lush vegetation period near the equator, so the observed annual 95th percentiles will be smaller than the real annual 95th percentiles. Therefore, it is needed to modify this bias before applying the percentiles to subsequent applications such as Earth's surface change analysis.

With the rapid emergence of massive remote sensing data, cloud computing technology came into being. In particular, since the launch of the Google Earth Engine (GEE), with its peta-byte-level distributed storage and high-performance cloud computing capabilities [35], [36], [37], the cloud platform brings great convenience to large-scale remote sensing applications. Remote sensing applications based on medium- and high-resolution long-time-series datasets are no longer limited by physical storage space and computing resources. More and more researchers are focusing on national and global studies. The cloud platform has served global change research, including global forest change, land surface water resource change, land use change, natural disaster mapping, and climate change [32], [38], [39]. At the same time, GEE's powerful data processing and computing capabilities bring the possibility to analyze the uncertainty quantitatively and to modify the bias in the Landsat annual percentile caused by insufficient observations in large regions.

In this article, we focus on percentile and propose a Statistical Time-series biAs Modification Model (STAMM) that can generate real Landsat percentiles using actual clear Landsat observation and reconstructed continuous daily dataset. Herein, the real Landsat percentile means Landsat percentile based on sufficient observations. STAMM is designed to provide consistent Landsat percentiles that can be applied to large-region or even global studies with the support of GEE. Meanwhile, STAMM can quantitatively evaluate the uncertainties in studies such as land cover mapping, change detection, and vegetation variation using the original Landsat percentile data. To obtain per-pixel real Landsat percentiles, our objectives are given as follows:

1) developing a Landsat percentile bias modification model based on the actual clear Landsat observation and continuous daily time-series dataset;
2) analyzing the biases of Landsat percentile due to insufficient Landsat observations;
3) modifying the Landsat percentile biases and analyzing its accuracy by taking the percentiles from reconstructed MODIS and Sentinel-2 time-series datasets as references.

II. Data and Materials

A. Landsat Surface Reflectance Products

In this study, we selected the most used vegetation index, normalized difference vegetation index (NDVI), as an example, and analyzed and modified the biases in 30-m Landsat 50th, 75th, and 95th percentile NDVI due to insufficient observations. Landsat NDVI was calculated based on the Landsat Collection 1 Tier 1 dataset, which consists of data products from 1972 to 2021 and was generated from Landsat 8 Operational Land Imager (OLI)/Thermal Infrared Sensor (TIRS), Landsat 7 ETM+, Landsat 4-5 Thematic Mapper (TM), and

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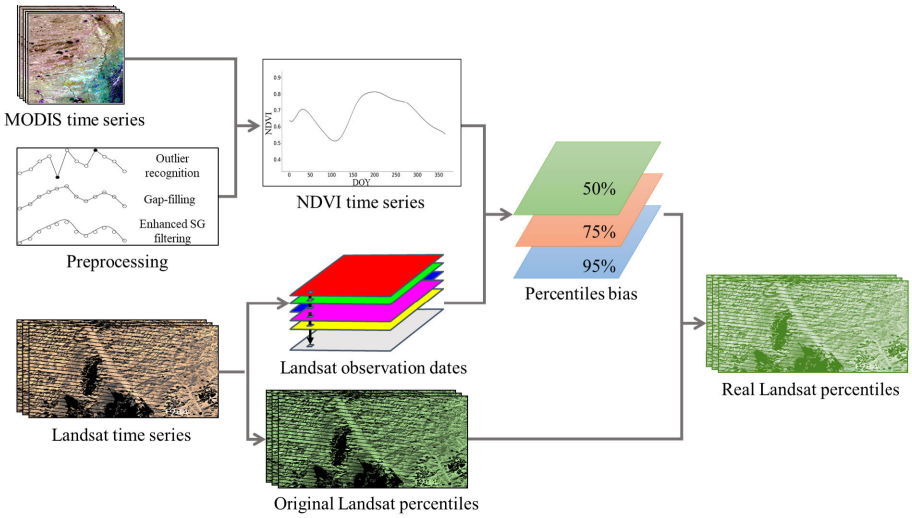


Fig. 3. Process of Landsat NDVI percentiles modification based on reconstructed continuous MODIS time series.

Landsat 1-5 Multispectral Scanner (MSS) instruments. Data in Tier 1 meet formal geometric and radiometric quality criteria. The radiometric performance of each Landsat sensor has been characterized and quantified at the instrument level and was critical to time-series analysis. The image-to-image registration accuracy of ≤12-m root-mean-square error (RMSE) was met, which was eligible for data "stacking" for time-series analysis [40]. Note that Landsat 4, 5, and 7 surface reflectance data are generated using the Landsat Ecosystem Disturbance Adaptive Processing System (LEDAPS; version 3.4.0) algorithm [41], while Landsat 8 surface reflectance data are generated using the Land Surface Reflectance Code (LaSRC; version 1.5.0), which makes use of the coastal aerosol band to perform aerosol inversion tests and uses auxiliary climate data from MODIS and a unique radiative transfer model [42].

B. MODIS Surface Reflectance Products

To analyze and modify the biases in the Landsat NDVI percentile, we used the eight-day MODIS surface reflectance product (i.e., MOD09Q1 version 6) to construct a continuous daily NDVI time-series dataset. MOD09Q1 provides an estimate of the surface spectral reflectance of Terra MODIS Red and Near-InfraRed (NIR) bands in 250-m spatial resolution, which has corrected for atmospheric conditions such as gases, aerosols, and Rayleigh scattering based on the Second Simulation of a Satellite Signal (6S) lookup table (LUT) [43]. For each MODIS pixel, only one reflectance value was selected from all the acquisitions within the eight-day composite period. The criteria for the choice of each pixel include cloud and solar zenith angles. When several acquisitions met the criteria, the pixel with the minimum blue value was used [44].

C. Sentinel Surface Reflectance Products

To analyze the utility of STAMM, we took the NDVI percentiles from the 10-m Sentinel-2 Level-2A surface reflectance product as the reference for comparison. The Level-2A product corrected for atmospheric effects through the MACCS-ATCOR Joint Algorithm (MAJA) software developed in coordination between the Centre National d'Etudes Spatiales (CNES) Centre d'Etudes Spatiales de la Biosphère (CESBIO) and Deutsches Zentrum für Luft- und Raumfahrt (DLR). This processor uses multitemporal information to detect clouds (and cloud shadows) and to estimate the optical properties of the atmosphere [45]. For this sensor, the absolute radiometric accuracy of the Level-1 product is 3% and the multitemporal calibration is 1%. The absolute geolocation uncertainty of the Level-1 product is about 12.5 m, and the uncertainty of the multitemporal registration, as well as the multispectral registration (for any two spectral bands), is all about 0.3 pixels, including compensation for the effects of terrain height variation (https://sentinel.esa.int/web/sentinel/technical-guides/sentinel-2-msi/performance). To compare with the Landsat NDVI percentiles, we resampled the 10-m Sentinel NDVI into 30-m resolution by simple average.

III. Theory and Methodology

The modification model for Landsat percentiles biases, STAMM, proposed in this study mainly depends on the difference between the real percentiles (i.e., percentiles from the reconstructed continuous daily NDVI time series) and the Landsat-like percentiles (i.e., percentiles from the NDVI time series that only contain observations when clear Landsat observations are available). STAMM consists of two steps: 1) estimate the bias degree with the reconstructed continuous daily NDVI time-series and clear Landsat observation dates and 2) modify the original percentiles from the actual clear Landsat reflectance based on the bias degree (Fig. 3). Since the Landsat and the Terra MODIS follow the same orbit, similar equator crossing times (i.e., differ by only about 30 min), and each Landsat multispectral band has a MODIS band with a similar bandwidth [46], [47], herein, we used MODIS to construct daily NDVI time-series dataset, considering that it has a higher temporal resolution. In this study, we assumed that the impact of insufficient observations on Landsat NDVI percentiles has the same degree (i.e., the relative difference in percentage format) as that on MODIS NDVI percentiles.

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By analyzing the relative difference between the MODIS real percentiles and simulated Landsat-like percentiles, the bias in the Landsat percentiles from the actual clear observations can be quantified and modified in equal proportions.

A. Continuous Daily MODIS Time-Series Construction

The continuous daily MODIS NDVI time series was constructed based on the NDVI calculated from MOD09Q1 [44]. Considering that the MODIS NDVI itself may also have problems of data missing due to poor quality, NDVI was optimized through the following four steps: 1) quality control; 2) outlier recognition; 3) gap filling; and 4) enhanced SG filtering. Quality control mainly used the quality assurance layers that come with MOD09Q1 products to remove clouds, cloud shadows, and other pixels marked as poor quality. Outliers in annual time series were recognized and removed since they always exhibit sudden increases or decreases. Herein, the outlier is defined based on two criteria: 1) its value is greater/lower than the mean of a moving window (i.e., 16 days in this study) plus/minus (i.e., sudden increases/decreases) twice the standard deviation of the points in the moving window [48] and 2) the absolute relative differences between its value and the two values before and after are all greater than 30% [49]. After removing the unavailable observations, a series gap filling algorithms and the enhanced Savitzky–Golay (SG) filtering were used to reconstruct and smooth the MODIS NDVI time series.

To obtain continuous daily MODIS NDVI, a Continuous Change Tracker Model (CCTM) for remote sensing time-series reconstruction was involved [50]. The CCTM includes components of gradual, periodic, and abrupt variations to capture the common surface change types [51]: 1) uniform variation; 2) decelerated variation; 3) accelerated variation, which is always caused by deforestation, floods, fire, insects, urbanization, and so on; and 4) periodic variation caused by seasonality, climate variability, vegetation growth, and so on. Since the seasonal trend of the MODIS NDVI time series may be quite different, in this study, we used the CCTM to fit each season. The CCTM for a single season is shown as follows:

$$\begin{aligned}\widehat{\rho}(s,x)_{\text{simple}} = a_0 \times \frac{\max - x}{\max - \min} + b_0 + a_1 \\ \times \cos\left(\frac{2\pi(x-\min)}{\max-\min}\right) + b_1 \\ \times \sin\left(\frac{2\pi(x-\min)}{\max-\min}\right) + a_2 \\ \times \left(\frac{e^{\frac{x-\min}{\max-\min}}-1}{e-1} \times 2 - 1\right) + a_3 \\ \times \ln\left(e^{-1} + \frac{x-\min}{\max-\min} \times \left(e - e^{-1}\right)\right) \quad (1)\end{aligned}$$

where x is the observation date in day of year (DOY) format; max is the maximum DOY of a single seasonal interval; min is the minimum DOY of a single seasonal interval; s represents the season (i.e., $s = 1, 2, 3, 4$); and a_0, b_0, a_1, b_1, a_2, and a_3 are the coefficients of different variation models. In this way, the continuous daily MODIS NDVI time series in a year was recorded as 24 coefficients per pixel (six coefficients of different variation models $\times$ four seasons) and could easily calculate MODIS NDVI at any DOY [50].

B. Landsat-Like Percentiles Simulation

The Landsat-like percentiles ($P_{k,\text{discrete}}$) were simulated by extracting the MODIS NDVI ($\text{NDVI}_{\text{modis}}(s,x)$) with the actual Landsat observation DOYs in a year (d_l) and counting its annual percentile values

$$P_{k,\text{discrete}} = f_k(\text{NDVI}_{\text{modis}}(s, d_l)). \quad (2)$$

In (2), f_k is the function for percentile statistic, k is 5, $10, \ldots, 95$. s for the season and d_l for the DOY are associated. To obtain the actual Landsat observation DOYs, we retrieved the Landsat surface reflectance via GEE in a year and carried out quality control. Pixels contaminated by clouds, cloud shadows, and snow or influenced by other problems were marked as invalid. For each Landsat pixel, we recorded its actual valid observation DOYs and extracted the corresponding MODIS NDVI based on CCTM. In this study, we simulated and analyzed the 50th, 75th, and 95th Landsat-like NDVI percentiles (i.e., $P_{50,\text{discrete}}$, $P_{75,\text{discrete}}$, and $P_{95,\text{discrete}}$).

C. Landsat Percentiles Modification

Based on the continuous daily MODIS NDVI time series, we can calculate the reference 50th, 75th, and 95th percentile values that contain sufficient surface change information. Considering that MODIS and Landsat have similar Red (MODIS 620–670 nm and Landsat 8 630–690 nm) and NIR (MODIS 841–876 nm and Landsat 8 850–880 nm) band settings when the two have the same insufficient observation distribution, the bias level of the statistically obtained percentile values should be equivalent. We compared and analyzed the differences between MODIS and Landsat NDVI in Section V-B through experiments. The difference between the simulated Landsat-like percentile values ($P_{k,\text{discrete}}$) and the percentiles with continuous observations ($P_{k,\text{continuous}} = f_k(\text{NDVI}_{\text{modis}}(s,x))$, $s \in [1,4]$, $x \in [1365/366]$) represents the bias in MODIS NDVI percentiles due to insufficient observations ($\text{bias}_{k,\text{absolute}}$)

$$\text{bias}_{k,\text{absolute}} = P_{k,\text{continuous}} - P_{k,\text{discrete}} \quad (3)$$

$$\text{bias}_{k,\text{relative}} = \frac{\text{bias}_{k,\text{absolute}}}{P_{k,\text{discrete}}} \quad (4)$$

where $\text{bias}_{k,\text{relative}}$ is the relative bias in Landsat kth percentiles. Meanwhile, the bias in Landsat percentile values was calculated as (5) and it can be modified based on (6)

$$P_{k,\text{bias}} = P_{k,\text{landsat}} \times \text{bias}_{k,\text{relative}} \quad (5)$$

$$P_{k,\text{modified}} = P_{k,\text{landsat}} \times (1 + \text{bias}_{k,\text{relative}}) \quad (6)$$

where $P_{k,\text{landsat}}$ is the original Landsat kth percentile values in a year, $P_{k,\text{bias}}$ is the absolute bias in Landsat kth percentiles due to insufficient observations, and $P_{k,\text{modified}}$ is the modified Landsat kth percentiles.

In this study, we produced the $\text{bias}_{k,\text{relative}}$ and $P_{k,\text{bias}}$ products for Landsat NDVI 50th, 75th, and 95th percentiles

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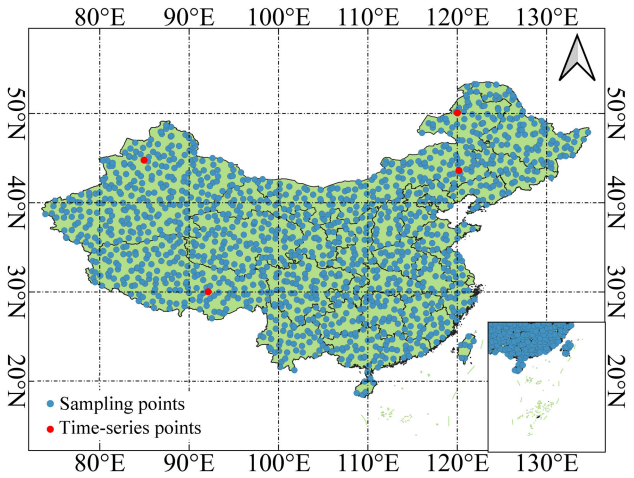


Fig. 4. Distribution of the sampling points distribution. The blue points are used for verification of space dimensions; the red points are used for time-series analysis.

in China from 1990 to 2020, year by year, with the help of MODIS. For the year before 2000 when MODIS product was not available, the reconstructed continuous MODIS NDVI in 2001 was used.

D. Evaluation of the STAMM

In this study, we show the change of Landsat NDVI percentiles before and after modification in both time and space dimension and evaluates the accuracy of the modified results in space dimension.

First, we compared the differences between Landsat NDVI 75th percentiles before and after modification from 2000 to 2020. Referring to the distribution of relative biases of Landsat NDVI 75th percentiles during this period, we selected four typical pixels for analysis. The relative biases of these pixels (red points in Fig. 4) are: 1) always positive (the southern part of the Qinghai–Tibet Plateau; 92.00°E, 29.99°N); 2) always negative (the northeast of Inner Mongolia; 119.99°E, 49.99°N); and 3) changes with time (the eastern of Inner Mongolia, 120.40°E, 43.56°N; and north of Xinjiang, 85.31°E, 45.11°N). The latitudes and longitudes in parentheses indicate the center position of each area, and the spatial range of the area is 300 × 300 m.

Furthermore, we randomly selected verification points on all Landsat's tiles covering China (blue points in Fig. 4) for evaluation. Then, we only kept the points that both Landsat and Sentinel had clear observations in 2019. Finally, 1377 points met the requirements. By constructing a linear regression model [see (7)] between Landsat ($NDVI_{Landsat}$) and Sentinel ($NDVI_{sentinel}$) NDVI on these verification points, we can simply get the transform coefficients (c and d) to weaken the spectral difference between these two datasets

$$NDVI_{Landsat} = c \times NDVI_{sentinel} + d. \tag{7}$$

We applied the coefficients c and d to the original $NDVI_{sentinel}$ and obtained the transformed Sentinel NDVI. Then, the transformed Sentinel NDVI was reconstructed into a continuous daily time series with CCTM. Finally, the reference Sentinel 75th percentiles were calculated from the reconstructed continuous daily transformed Sentinel NDVI.

IV. Results and Analysis

A. Spatial Distribution of the Landsat Percentile Biases

The relative and absolute biases in Landsat NDVI 75th percentiles in China were analyzed and their spatial distribution in 2015 is shown in Fig. 5. According to (3), the positive values (i.e., blue color in Fig. 5) represent the original Landsat percentiles underestimating the real ones, while the negative values (i.e., red color in Fig. 5) represent overestimation. As we can see, the biases of Landsat NDVI percentiles are unevenly distributed throughout China. In general, Landsat percentiles over both northeast and northwest China overestimated the real NDVI 75th percentiles [the relative overestimation is about 10%–30% in Fig. 5(a); the absolute difference is about −0.1 in Fig. 5(b)], while in the southwest of China, it has an underestimation problem [the relative underestimation is about 10% in Fig. 5(a); the absolute difference is about 0.1 in Fig. 5(b)]. The eastern part of the Qinghai–Tibet Plateau and the middle and lower reaches of the Yangtze River are more complex, with obvious overestimation and underestimation at the same time.

The distribution of biases has a clear correlation with the number of clear Landsat observations (#L) during the same period (Fig. 6). In the northeast and northwest of China, there are more observations in the summer [#L: 5–10; Fig. 6(b)] and autumn [#L: 5–15; Fig. 6(c)] than in spring [#L: 1–10; Fig. 6(a)] and winter [#L: 0–5; Fig. 6(d)]. Especially during the whole winter, some pixels only have one clear observation due to the snowing issue. Such biased observation leads to relatively more high-value data and insufficient low-value data participating in percentile statistics, which eventually leads to overestimation of percentiles. In the southwest of China, more observations appear in spring and winter, while fewer observations are in summer and autumn due to rain issues, and this makes the original Landsat 75th percentiles underestimate the real ones. Although there is abundant rainfall in both the eastern Qinghai–Tibet Plateau and the middle and lower reaches of the Yangtze River [52], the differences in the number of observations during the four seasons of the Yangtze River estuary are not as large as those in the eastern part of the Qinghai–Tibet Plateau (i.e., the relative bias is not that obvious).

B. Temporal Variation of the Landsat Percentile Biases

Fig. 7 shows the distribution of relative biases of Landsat NDVI 75th percentiles from 2000 to 2019. Like the results in 2015 [Fig. 5(a)], Landsat overestimated the 75th percentiles in the northeast and northwest of China during the past 20 years [Fig. 7(a)]. As for the east of Inner Mongolia [Fig. 7(b)], the biases of Landsat percentile in the past 20 years showed a change of underestimation, overestimation, and then underestimation, while for the north of Xinjiang province [Fig. 7(c)], the relative bias increased year by year. Considering that Landsat's orbit and tile overlaps are fixed, the impact of meteorological factors on data quality is long term and gradually changes in space. Also, such fluctuations in local areas are related to changes in land cover types caused by human activities (Section V-D discusses this in detail). However,

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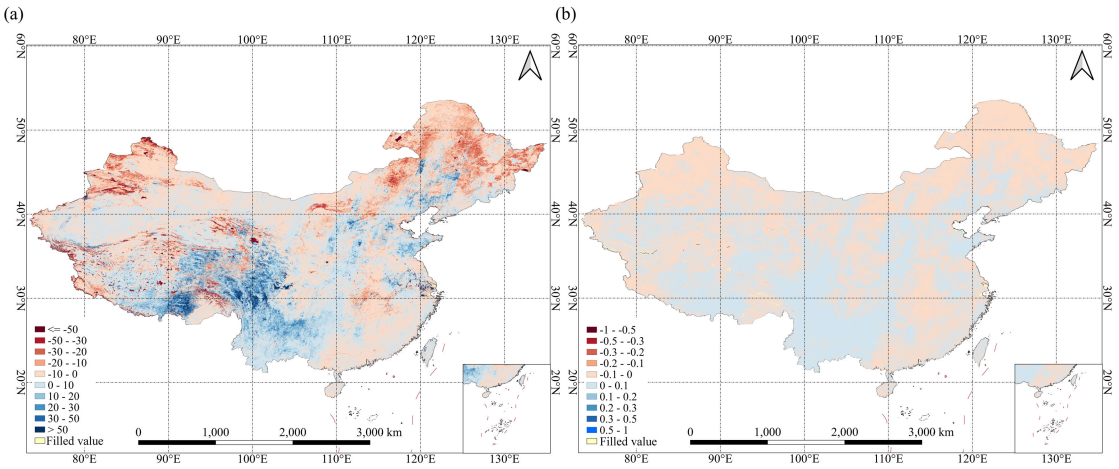


Fig. 5. Distribution of relative and absolute biases of NDVI 75th percentiles in 2015. (a) Relative biases. (b) Absolute biases. The positive values represent underestimation, while the negatives represent overestimation.

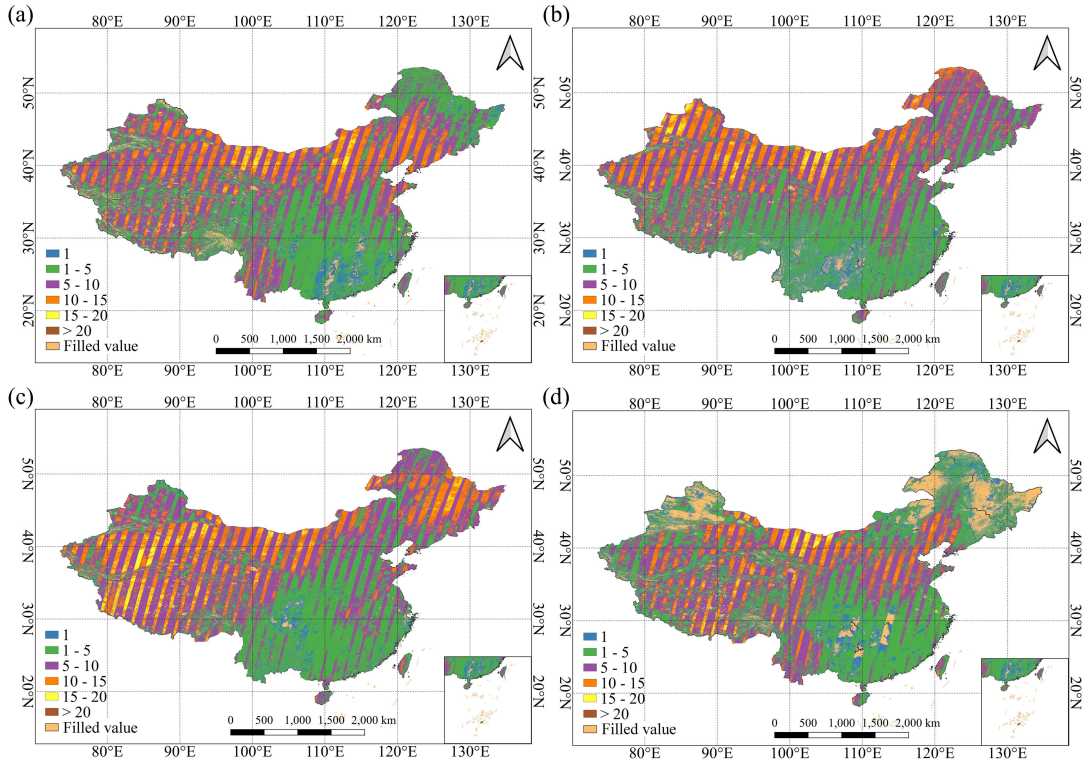


Fig. 6. Number of Landsat observations in 2015 in China. (a)–(d) Spatial distribution in spring, summer, autumn, and winter, respectively.

in the region of the eastern Qinghai–Tibet Plateau, it faces a very serious underestimation problem [Fig. 7(d)]. This is mainly because of the relatively low latitude and the obvious difference in terrain, so the east of the Qinghai–Tibet Plateau is always cloudy and rainy. During the summer, when vegetation here is in the lushest period, the abundant rainfall results in fewer clear Landsat observations. Therefore, there are fewer high values involved in percentile statistics, resulting in the underestimation problem. In general, during the past 20 years, Landsat NDVI 75th percentile shows an underestimated issue in most of China [Fig. 7(a)]. However, this problem has eased in the last five years (i.e., 2015–2019; Fig. A1 in the Appendix presents the five-year averaged distribution of relative biases of Landsat NDVI 75th percentiles for whole China).

C. Accuracy Analysis of the STAMM

1) Time-Series Comparison: The Landsat NDVI 75th percentiles after modification solve the always-underestimated [Fig. 8(a); the yellow circle in Fig. 7(d)] and always-overestimated [Fig. 8(b); the very top yellow circle in Fig. 7(b)] problem of the original percentiles during 2000–2019 efficiently. As for eastern Inner Mongolia [the lower yellow circle in Fig. 7(b)], relative biases of Landsat NDVI 75th percentiles vary year by year. From 2010 to 2014, there was an obvious overestimation area (red color) in eastern Inner Mongolia on the 120°E longitude line [Fig. 7(b3)]. In Fig. 8(c), the red curve (i.e., Landsat NDVI 75th percentiles after modification) during this period is lower than the blue curve representing the original percentiles. The selected area

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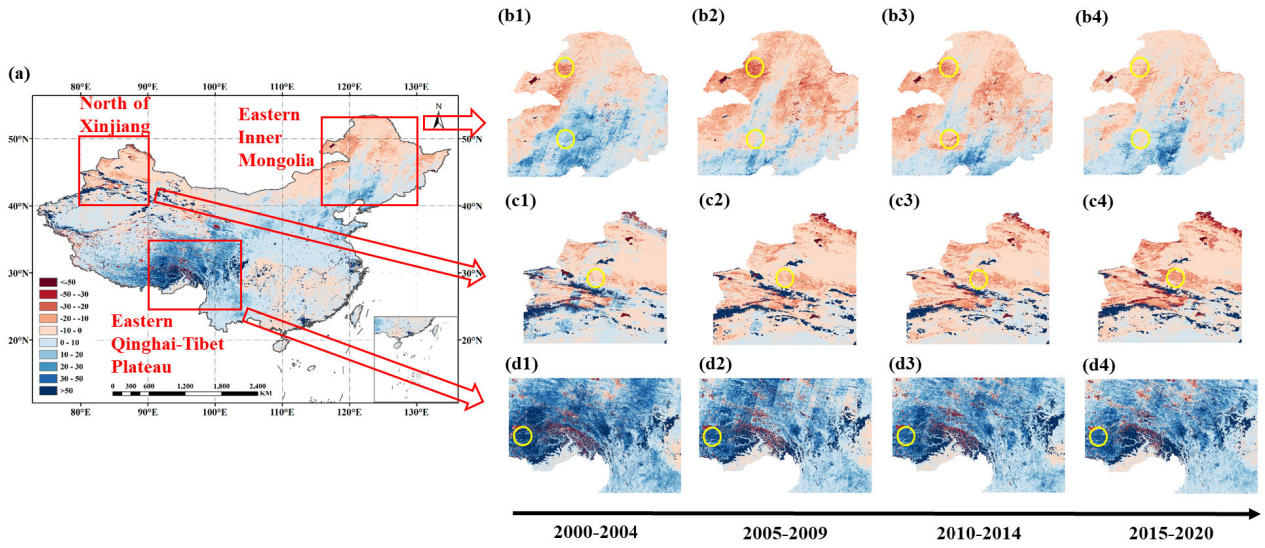


Fig. 7. Distribution of relative biases of Landsat NDVI 75th percentiles from 2000 to 2019. (a) Average relative biases during 2000–2019 in China. (b) Relative biases of percentiles in eastern Inner Mongolia (40°N–55°N, 115°E–130°E). (c) North of Xinjiang province (40°N–50°N, 80°E–90°E). (d) Eastern Qinghai–Tibet Plateau (25°N–35°N, 90°E–105°E). The average relative biases during (1) 2000–2004, (2) 2005–2009, (3) 2010–2014, and (4) 2015–2019. The four yellow circles mark the four areas analyzed in Fig. 8. The positive values represent underestimation, while the negatives represent overestimation.

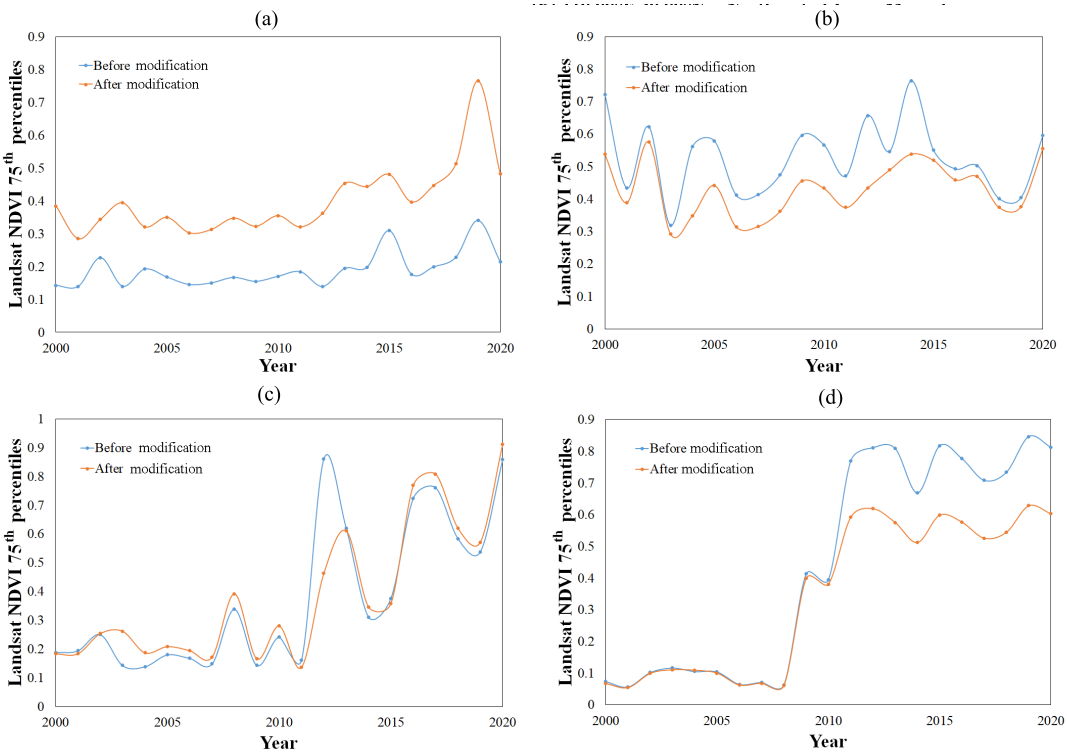


Fig. 8. Time series of Landsat NDVI 75th percentiles before and after modification from 2000 to 2020. (a) Eastern Qinghai–Tibet Plateau. (b) Northeast of Inner Mongolia. (c) Eastern Inner Mongolia. (d) North of Xinjiang province. Each subfigure comes from a 300 × 300 m area. The latitude/longitude is the central location for each area.

in the north of Xinjiang province [Fig. 8(d)] was bare land before 2010, and therefore, the yearly NDVI percentiles are below 0.1. Since there is almost no NDVI variation during a year for bare land, there was no difference between modified NDVI percentiles and the original ones. After it was gradually reclaimed as cropland, affected by snowfall in winter, which is the same as the northeast of Inner Mongolia [Fig. 8(b)], it was easy to lack low-value observations and overestimate the yearly percentiles. The modified percentile effectively mitigates the problem of overestimation [Fig. 8(d)].

2) Compared With Sentinel Percentiles: Taking the 75th percentiles of Sentinel NDVI as a reference, Fig. 9 compares Landsat NDVI percentiles with Sentinel. After modification, the consistency (R^2) between NDVI percentiles based on Landsat and Sentinel increases from 0.974 to 0.987, while the difference (RMSE) decreases from 0.065 to 0.045. Since the Sentinel NDVI percentile was derived from the reconstructed continuous Sentinel time-series data and the reconstruction process is independent of the MODIS reconstruction used to modify Landsat, these results prove that the modification

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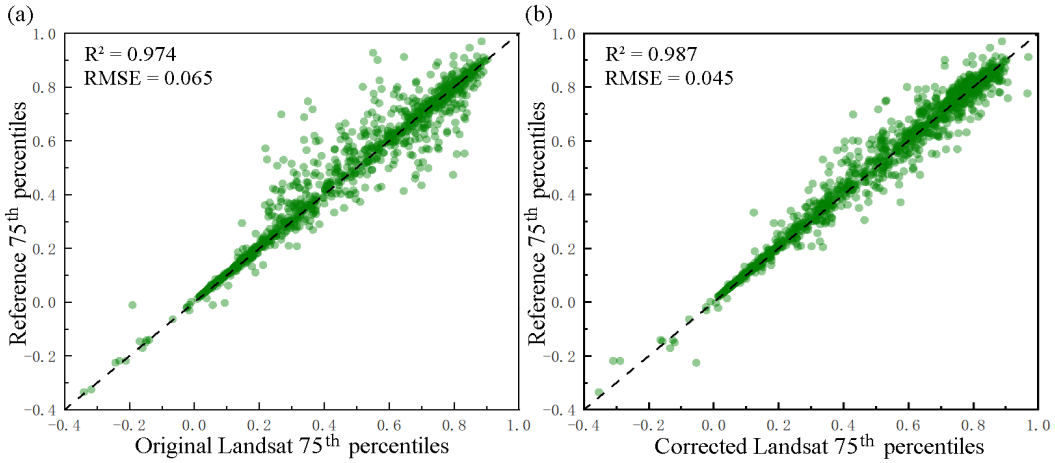


Fig. 9. Comparison between Sentinel NDVI 75th percentiles and the corresponding Landsat percentiles (a) before and (b) after modification.

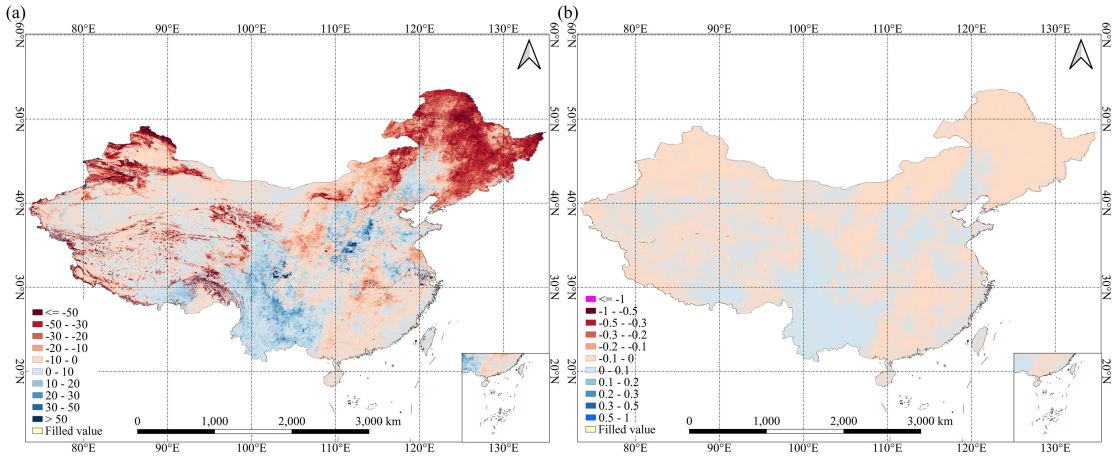


Fig. 10. Distribution of relative biases of NDVI 50th percentiles in 2015. (a) Relative biases. (b) Absolute biases. The positive values represent underestimation, while the negatives represent overestimation.

model, STAMM, provided in this study deed improves the accuracy of Landsat percentiles.

D. Comparison Among Different NDVI Percentiles

Like Fig. 5, Figs. 10 and 11 show the spatial distribution of the relative biases in Landsat NDVI 50th and 95th percentiles, respectively. As we can see, the overestimation issue over the northeast and northwest of China is more serious for the 50th percentiles [the relative bias is about 30%–50%; Fig. 10(a)] than for the 75th percentiles [about 10%–30%; Fig. 5(a)]. Moreover, there is almost no overestimation issue for 95th percentiles in the same areas [Fig. 11(a)]. As for the eastern part of the Qinghai–Tibet Plateau, due to missing data throughout the year, all three percentiles are underestimated. In general, there is a large number of areas that face the overestimation of the 50th percentile, and only a few areas have the problem of seriously underestimating the 50th percentile (i.e., >20%). In the northeast of China, the relative bias has exceeded 30%. However, since the absolute 50th percentile is overall small, the absolute bias is less than 0.1. The absolute bias of the 50th percentile all over China is between −0.1 and 0.1, and there are slightly more overestimated areas [Fig. 10(b)]. The 95th percentile has the smallest bias all over China, and the biases of both overestimation and underestimation are less than 10%, with the absolute bias less than 0.1. At the same time, the 95th percentile in most parts of China is slightly underestimated [Fig. 11(b)]. The reason why the relative bias of the 95th percentile is small is that the absolute value of the 95th percentile is relatively large.

V. Discussion

A. Difference Between Different NDVI Percentiles

The spatial distribution of the relative biases in Landsat NDVI 50th, 75th, and 95th percentiles varies greatly (Figs. 5(a) and 10). Research has reached the same conclusion by reconstructing the Landsat time series in a local area and comparing the different percentiles based on the reconstructed series and the original data [34]. This is because if the data used for statistics are biased and there is much more data in the high-value period than in the low-value period, the statistical impact on the high-value indicators (e.g., maximum and 95th percentile) will be small, while, due to the lack of statistical data in the low-value period, the bias for indicators representing the median (e.g., 50th percentile) will be large. It is conceivable that if these data are used for the statistics of the minimum value (e.g., the 5th and 10th percentiles), there

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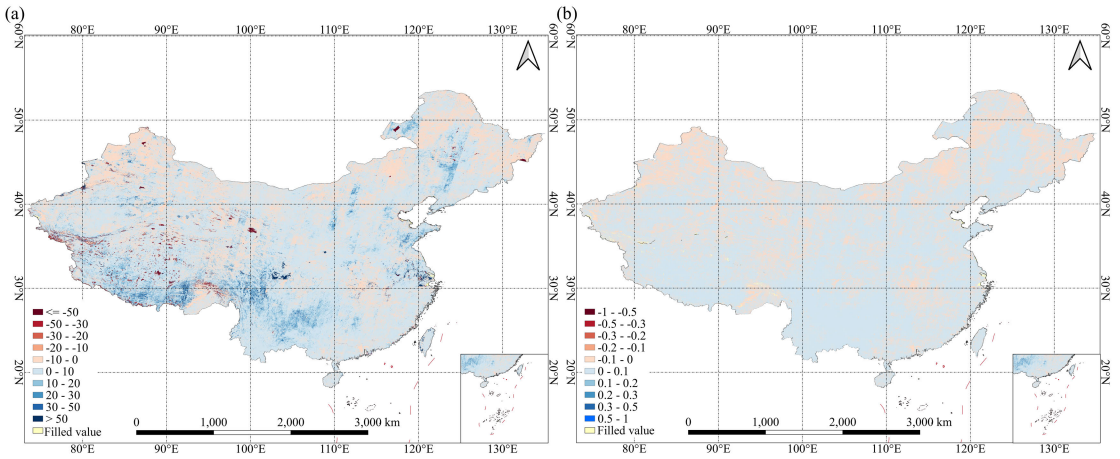


Fig. 11. Distribution of relative biases of NDVI 95th percentiles in 2015. (a) Relative biases. (b) Absolute biases. The positive values represent underestimation, while the negatives represent overestimation.

will be a large bias. As for the underestimation issue over the southwest of China, due to the lack of data in the high-value period, there is not much difference among the 50th, 75th, and 95th percentiles. These three indicators cannot accurately obtain high-value results. In general, after excluding outliers, the 95th percentile values are more stable for insufficient observations. For most of China, the deviation of the 95th percentile is less than 10%.

B. Feasibility Analysis of STAMM Using MODIS

In this study, we provided a model to modify the Landsat percentile bias caused by insufficient observation based on the reconstructed continuous MODIS time series. MODIS has long been used to make up for the low observation frequency of Landsat [11], [12], [13], [14], [15]. Using the MODIS time series to modify the bias in Landsat statistical metrics can keep the phenology information, at the same time, without losing the simplicity and efficiency of statistical metrics. The effective observation frequency of MODIS itself is much higher than that of Landsat, which is more conducive to time-series data reconstruction. MODIS has a lower spatial resolution than Landsat and uses fewer resources during reconstruction, which allows the STAMM to be used for large-region study.

The consistency between MODIS NDVI and Landsat was evaluated to verify the feasibility of STAMM based on the 1377 verification points in China (Fig. 4). The results proved that it is feasible to modify the Landsat percentile bias using MODIS time-series data. As each MODIS pixel (250 m) covers about 81 Landsat pixels (30 m), the consistencies (R^2) between MODIS NDVI and each Landsat NDVI in the MODIS pixel area were calculated and averaged to represent the consistency between MODIS and these 81 Landsat NDVIs. Fig. 12 shows the cumulative distribution histogram of the time-series consistency between MODIS NDVI and Landsat at all sampling points from 2001 to 2020. Nearly, 70% of the sampling points have a consistency above 0.6, but there are also less than 5% of the sampling points, whose consistency is negative. This should be related to the assignment of critical pixels in the spatial matching process between MODIS and Landsat. Therefore, our experiment also proves that NDVI

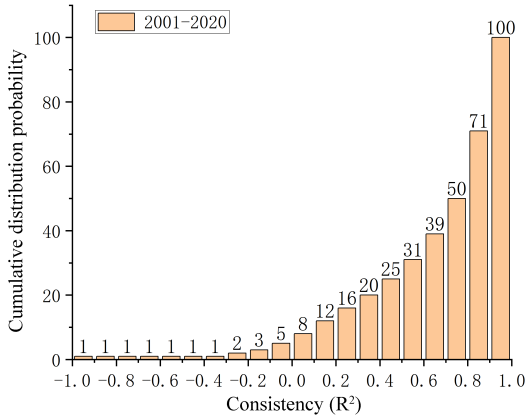


Fig. 12. Histogram of the cumulative distribution of consistency between MODIS NDVI and Landsat.

from MODIS and Landsat has good consistency, and it is feasible to use MODIS NDVI time series to modify the bias in Landsat. This conclusion is consistent with existing research [11], [12], [13], [14], [15].

C. Uncertainty Analysis of Using 2001 MODIS to Modify the Landsat Percentiles Bias Before 2000

Landsat has a longer time series than MODIS. Usually, when needed to characterize vegetation dynamics before MODIS (i.e., before 2000), researchers would like to fuse MODIS, advanced very high resolution radiometer (AVHRR), or satellite pour l'observation de la terre (SPOT) data [53], [54], [55]. However, due to the difference in the band settings, this will lead to uncertainties in the estimated NDVI [56], [57]. In this study, for the years from 1990 to 1999, the MODIS data of 2001 were used since this is the first year that is fully covered by MODIS. To analyze the uncertainty about this approach, we took 2009 as an example analysis. Landsat NDVI 75th percentiles in 2009 were modified using MODIS from 2009 (marked as Landsat_m2009) and 2019 (marked as Landsat_m2019). Fig. 13 shows the difference between Landsat_m2009 and Landsat_m2019 (i.e., Landsat_m2019 and Landsat_m2009). Results show that both the overestimated and underestimated areas of Landsat_m2009 and Landsat_m2019

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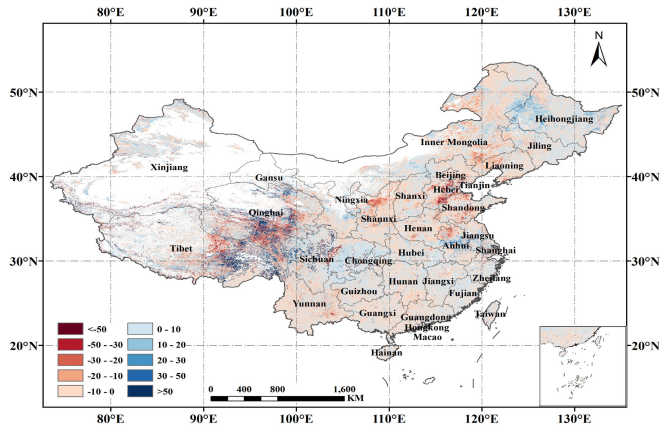


Fig. 13. Difference of relative biases between 2009 Landsat NDVI 75th percentiles modified by 2009 and 2019 MODIS.

are very concentrated. Among them, the eastern Qinghai–Tibet Plateau and the central part of China (i.e., Sichuan Province) have both overestimated and underestimated areas, and quite a few of these areas have differences of more than 30%. In addition, the overestimation bias in the east of China (i.e., the Beijing–Tianjin–Hebei region) is also relatively large, and the percentage of overestimation bias in a considerable part of the region exceeds 30%.

The feasibility of using the continuous MODIS time series in 2001 to modify the Landsat percentiles from 1990 to 1999 is that the land cover types do not change significantly. When the local land cover type changes, both the variation trend and the range of MODIS time-series data will change significantly. Therefore, in the future, based on land cover products, for areas with different land types in 1990–1999 and 2001, the nearest MODIS data of the same land type can be selected for modification. Considering that in STAMM, the continuous MODIS time-series data are only used to provide the real variation trend of the surface. Any high-frequency continuous surface observation can be used instead of MODIS to modify the bias of Landsat due to insufficient observations. However, considering the angle (i.e., the illumination direction and observation direction) and spectral differences among different satellites, necessary consistency processing is required.

D. Analysis of Influencing Factors of Relative Biases

The relative bias of statistical metrics evaluated and modified herein is caused by insufficient observations. However, the amount of the relative bias is affected by both land cover type and meteorological factors. In this study, we take NDVI as an example to analyze and modify the bias in yearly percentiles quantitatively based on STAMM. As for artificial land, bare land, and other areas where NDVI does not change much within a year, the relative bias is small. As for vegetation, especially for deciduous forests, grassland, and cropland, where NDVI has obvious seasonal variation, whether the limited observations can capture key phenological points directly affects the amount of the bias. Fig. 14(a) shows the land cover change from 2000 to 2019 in eastern Inner Mongolia [Figs. 7(b) and 8(c)], while Fig. 14(b) is for the north of Xinjiang province [Figs. 7(c) and 8(d)]. For eastern Inner Mongolia, before 2010, it was mainly grassland [Fig. 14(a1) and (a2)], with a small NDVI percentile and a small bias. When it was reclaimed as cropland [Fig. 14(a3) and (a4)], the NDVI percentiles increased significantly as well as the bias caused by insufficient observation [Fig. 8(c)]. For the north of Xinjiang province, the selected pixel was almost bare land before 2010 [Fig. 14(b1) and (b2)], there is nearly no bias in the yearly NDVI percentiles; however, when it became cropland [Fig. 14(b3) and (b4)], both the yearly NDVI percentiles and relative bias increase significantly [Fig. 8(d)].

Whether the key points of vegetation phenology can be captured is related to meteorological factors such as precipitation. Fig. 15 shows the relationship between the number of precipitation days and the relative bias over time. The Level-3 0.1° spatial resolution daily precipitation data from Global Precipitation Measurement (GPM) based on the Integrated Multi-satellitE Retrievals for GPM (IMERG) algorithm were used to evaluate the relationship between the relative bias of NDVI percentile and precipitation. The number of precipitation days was days with precipitation of more than 0.1 mm. It can be seen that the annual number of precipitation days has no obvious relationship with the relative bias of the year, but in the long time series, it can affect the amplitudes of the relative bias. For eastern Inner Mongolia, the number of precipitation days has been slowly decreasing in the past 20 years, and the relative bias has also slightly decreased as a whole. From 2012 to 2015, the number of precipitation days decreased for four consecutive years, and at the same time, the relative bias also reduced volatility [Fig. 15(a)]. As for the north of Xinjiang province, the number of precipitation days has not changed much during the past 20 years, the big jump in relative bias is mainly caused by the change in land cover type, and the amplitudes before and after 2010 do not change much [Fig. 15(b)].

E. Outlook for STAMM

STAMM reconstructs the daily MODIS to obtain the relative percentile biases due to insufficient observations. Compared with Landsat, MODIS has higher observation frequency but lower spatial resolution; therefore, the reconstruction of MODIS can have higher precision [50]. However, the advantage of Landsat in spatial resolution is irreplaceable by MODIS. Herein, we use MODIS to construct surface change trends and to evaluate and modify Landsat percentiles. Compared with the fusion compositing algorithms [11], [12], [13], [14], [15] and algorithms that reconstruct Landsat based on the time-series model [34], STAMM can save a lot of computing power and storage resources. First, due to their spatial resolution difference, the storage space required to reconstruct MODIS (500 m) data is about 1/17 of that of Landsat (30 m). Second, STAMM uses CCTM to reconstruct daily MODIS data (i.e., 365/366 values per pixel), and only 24 values need to be stored per pixel per year, which means that the storage space is about 1/15 of the daily product. Finally, we estimated the relative bias percentage with the help of the DOYs of actual clear Landsat, directly modified the annual percentiles of Landsat (i.e., no longer need to

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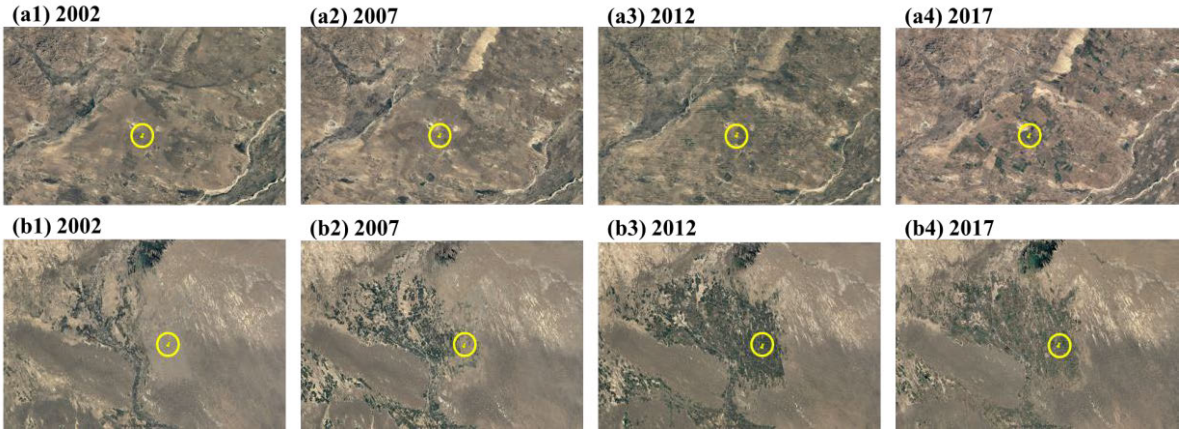


Fig. 14. Land cover change for (line a; 120.40°E 43.56°N) eastern Inner Mongolia and (line b; 85.31°E 45.11°N) north of Xinjiang from 2000 to 2019. All images were captured from Google Earth.

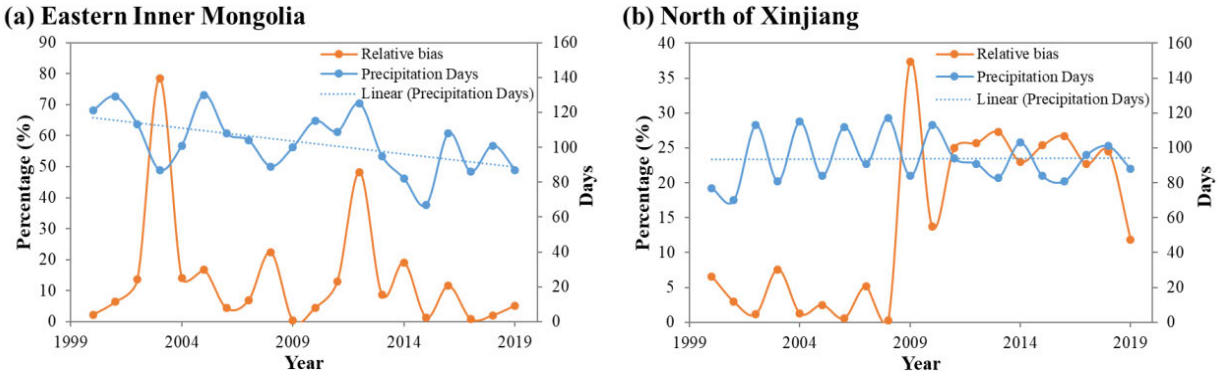


Fig. 15. Relationship between the relative bias and the number of precipitation days in a year for (a; 120.40°E 43.56°N) eastern Inner Mongolia and (b; 85.31°E 45.11°N) north of Xinjiang province from 2000 to 2019.

reconstruct the time series of Landsat). This greatly improves the modification efficiency.

STAMM is a model developed based on the GEE platform, which can efficiently call and process a large amount of remote sensing data and can realize the modification of large-region or even global data. Furthermore, other satellite data that can construct surface change trends can be used to replace MODIS. This means that STAMM has strong applicability. STAMM can also modify the statistical metric from Sentinel, Chinese GaoFen (GF) series, and other relatively low-frequency satellite data, which have insufficient observation issues. On the other hand, STAMM can quantitatively evaluate the uncertainties in studies using the original statistical metric from Sentinel, Chinese GF series, and other relatively low-frequency satellite data.

In this study, we produced the annual Landsat NDVI 50th, 75th, and 95th percentiles relative and absolute bias product from 1990 to 2020 in China for Landsat modification. The modified Landsat percentiles can help improve the quality of land cover classification and mapping accuracy [34]. STAMM makes the percentiles between years comparable and is of great significance for long-time-series change detection. At the same time, STAMM can provide technical support for the uncertainty analysis of existing research based on the original Landsat percentile. However, the impact of insufficient observations on Landsat NDVI percentiles comes from the reconstructed MODIS NDVI with CCTM. According to the research on CCTM, the difference (RMSE) between the reconstructed NDVI time series and the original is less than 0.01 [50, Fig. 8], while the consistency is over 0.98 [50, Fig. 9] for most land cover types. This uncertainty will also affect the modification of percentiles by STAMM, and the evaluation of the modification results based on reconstructed MODIS and Sentinel-2 NDVI time series in this study. Since CCTM can not only construct NDVI time series but also time–series curves of leaf area index (LAI), gross primary production (GPP), and so on [50], STAMM can also modify the biases/evaluate the uncertainty in their percentiles or other statistical metric.

VI. Conclusion

In this study, we proposed a statistical time-series biases modification model (STAMM) and applied it to modify the long-time-series Landsat percentiles that suffer insufficient observation issues. By focusing on percentile metrics, STAMM can generate real Landsat percentiles in large regions by using actual clear Landsat observation (i.e., both DOY and reflectance) and the reconstructed daily MODIS time series. First, STAMM estimates the relative bias in the percentage of yearly Landsat NDVI percentiles due to insufficient observations based on the actual clear Landsat DOYs and the reconstructed daily MODIS time series. Then, the original Landsat NDVI percentiles from the actual clear Landsat reflectance are modified by the relative bias to get the real

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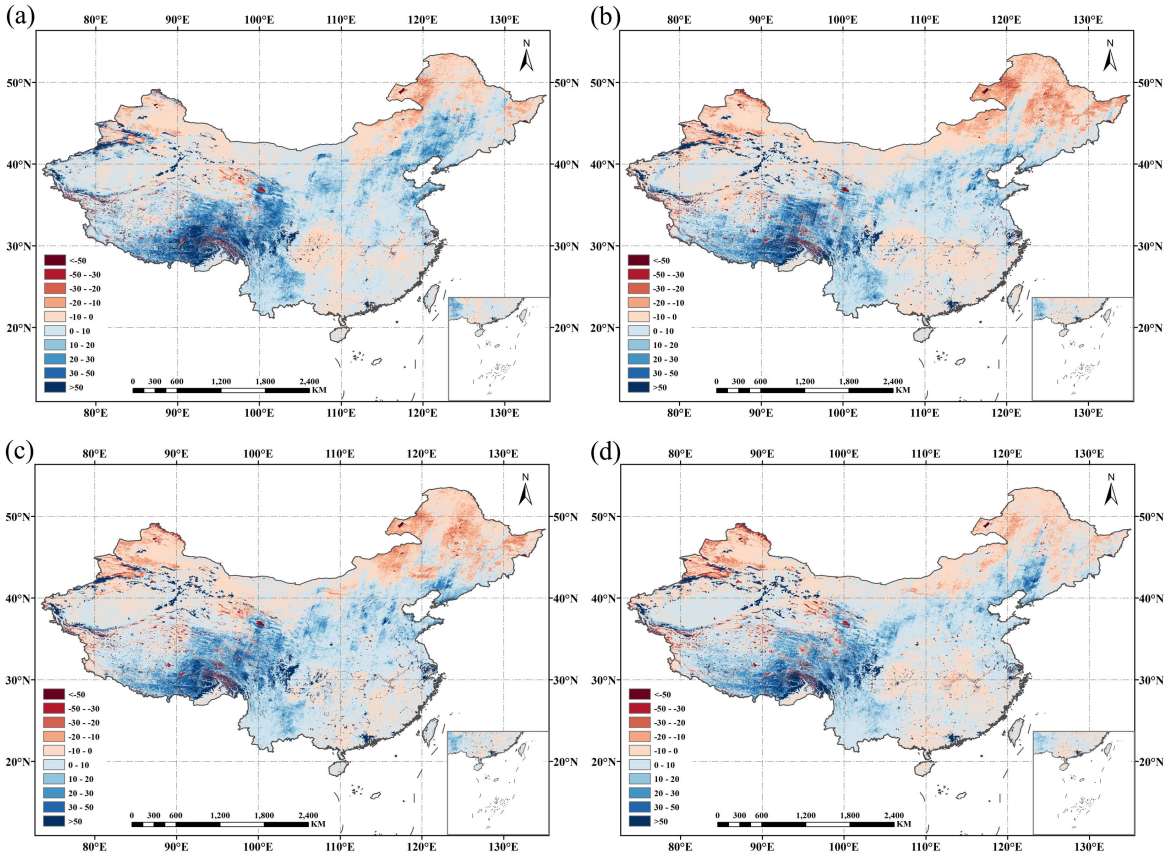


Fig. A1. Distribution of relative biases of Landsat NDVI 75th percentiles in China from 2000 to 2019. The average relative biases (a) during 2000–2004, (b) 2005–2009, (c) 2010–2014, and (d) 2015–2019. The positive values represent underestimation, while the negatives represent overestimation.

Landsat percentiles. Compared with the traditional method of reconstructing Landsat, STAMM can greatly save the storage required in the modification process and is more suitable for large-scale applications. During the modification, the continuous MODIS time series and the actual Landsat observation DOY are used to modify the biases of percentiles caused by insufficient Landsat observations. With the support of GEE, we produced the annual relative and absolute biases of Landsat NDVI 50th, 75th, and 95th percentiles from 1990 to 2020 in China and modified the corresponding percentiles.

By comparing with MODIS NDVI percentiles, the Landsat NDVI 75th percentiles over both the northeast and northwest of China overestimated about 10%–30% due to insufficient observation, while in the southwest of China, it underestimated about 10%. This issue has persisted for the past 20 years. However, it has eased in the last five years (i.e., 2015–2019). As for the different percentiles, the amount of bias depends on the characteristics of the data distribution. When there are many actual observations in the high-value area, the bias of the 95th percentile is relatively small and the bias of the 50th percentile bias is large. By comparing with the percentiles from the reconstructed and spectral normalized Sentinel NDVI time series, STAMM reduced the uncertainty of Landsat percentiles from 0.065 to 0.045, effectively improving the accuracy of Landsat percentiles. Therefore, STAMM can provide consistent and accurate Landsat percentiles regionally and even globally that are important to map land cover and derive interannual Earth's surface variation. Meanwhile, it can give a quantitative evaluation of the uncertainties when using the original Landsat percentile for land cover mapping.

APPENDIX

See Fig. A1.

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